

Mainul Islam

Researcher | Engineer | Robotics, Control & Machine Learning

Dhaka, Bangladesh | mainulislam@iut-dhaka.edu | [Google Scholar](#) | [LinkedIn](#) | [Portfolio](#) | [YouTube](#)

Research Interests

Full-stack field robotics, machine learning, computer vision for autonomous systems, unmanned ground vehicles and robotic manipulation, unmanned aerial vehicle, sensor fusion and state estimation, model predictive and learning-based control, and embedded robotic systems.

Education

B.Sc. in Electrical and Electronic Engineering (EEE)

Jan 2020 – Jun 2024

Islamic University of Technology (IUT), Gazipur, Bangladesh

CGPA: 3.87 / 4.00

Thesis — MonoSIM: Developed an open-source Software-in-the-Loop (SIL) framework for autonomous Ackermann vehicles using a monocular-vision, sliding-window lane-detection pipeline, and implemented and benchmarked Model Predictive Control (MPC) and PID strategies for path tracking within the simulation environment (published, IEEE ICARA 2026).

Publications

[Google Scholar link](#)

Peer-Reviewed Journal Article

- A. U. R. Adib, **M. Islam**, M. S. Abid, R. Ahshan, “A deep learning based framework for solar panel segmentation and fault classification enhanced with explainable AI,” *Solar Energy*, vol. 302, 2025. (Impact Factor: 6.6)
DOI: [10.1016/j.solener.2025.114058](https://doi.org/10.1016/j.solener.2025.114058)

Peer-Reviewed Conference Papers

- S. Rahman, N. Hasin, **M. Islam**, M. Z. A. Rony, G. Sarowar, “MonoSIM: An open-source SIL framework for Ackermann vehicular systems with monocular vision,” *IEEE 12th Intl. Conf. on Automation, Robotics and Applications (ICARA)*, 2026. (**B.Sc. Thesis**)
DOI: [10.1109/ICARA69401.2026.11480327](https://doi.org/10.1109/ICARA69401.2026.11480327)
- N. Hasin, M. A. R. Nishat, **M. Islam**, K. S. A. Hasan, A. Newaz, “A robust deep learning framework for Bangla license plate recognition using YOLO and vision-language OCR,” *IEEE Intl. Conf. on AI and Data Analytics (ICAD)*, 2026.
DOI: [arXiv:2603.10267](https://arxiv.org/abs/2603.10267)
- H. Haque, **M. Islam**, M. A. I. Alvy, M. S. Islam, M. A. Razzak, “Long-term energy demand analysis using machine learning algorithms: A case study in Bangladesh,” *IEEE ICECIE*, 2024.
DOI: [10.1109/ICECIE63774.2024.10815643](https://doi.org/10.1109/ICECIE63774.2024.10815643)
- M. S. Tanvir, F. A. Tonmoy, M. M. Islam, **M. Islam**, M. Hasan, “Optimizing master-slave broadcast and unicast communication in multi-node STM32 networks using UART and RS485 protocols,” *IEEE ICCIT*, 2024.
DOI: [10.1109/ICCIT64611.2024.11022430](https://doi.org/10.1109/ICCIT64611.2024.11022430)

Teaching Experience

Part-Time Lecturer, Department of EEE — Ahsanullah University of Science and Technology (AUST)
Jul 2025 – Jun 2026

- Instructed the **C++ Programming (EEE-1110)** and the **Digital Signal Processing (EEE-3218)** across two consecutive semesters, including lecturing, lab-manual preparation, and student assessment.

Professional Experience

Research and Development Engineer / Embedded Systems Engineer — Cybernetics Hi-Tech Solutions Pvt. Ltd. ([link](#)) Oct 2024 – Present

- **Autonomous Ground Vehicle (AGV):** Built a sensor-fusion system combining cameras, wheel odometry, and visual markers to help the robot track its own position accurately. Added LiDAR-based obstacle avoidance and combined navigation, localization, and motion control into one ROS2 system. Fixed a network bottleneck that was slowing down robot communication, and set up the system to start automatically on boot.
- **Man-Portable Teleoperated Robot:** Built the communication and control system for an 11-motor robot, including safety checks and automatic responses to faults or low battery. Added a backup wireless/fiber-optic link so the robot stays controllable if the main connection drops. Designed the motion control for a robotic arm and built a live video system with 5 cameras, a 3D model viewer, and remote camera control, all with minimal delay.
- **Toll-Booth Automation:** Built an automated system to detect vehicles, verify payment, and flag potential theft, using vehicle sensors and license-plate recognition. Also designed a custom circuit board for the hardware, with protection against tampering.

Technical Skills

Programming: Python, C/C++, MATLAB, Embedded C

ML / DL: PyTorch, TensorFlow, scikit-learn, YOLO, Explainable AI (LIME/SHAP), Vision-Language OCR

Robotics / Control: ROS2, Extended Kalman Filter (EKF), sensor fusion, inverse kinematics, MPC, PID, LiDAR/vision perception, AprilTag/vision-based localization

Embedded & Communication Protocols: CAN Bus, RS-485, UART, Modbus, STM32, PCB design, embedded systems, real-time distributed control

CAD / 3D Design: Autodesk Fusion, Onshape

Honors & Achievements

- **IRC 2024 — 6th Global:** International Rover Challenge, ABEX Best Science Team (Project Altair, 1st in Bangladesh).
- **ERC 2023 — 16th Global:** European Rover Challenge (Project Altair).
- **IRDC 2021 — 11th Global:** Indian Rover Design Challenge (Team Avijatrik, 1st in Bangladesh).
- **MATLAB Cody:** Global rank 117; 914 problems solved; 26 badges. ([Profile](#))

References

Prof. Dr. Golam Sarowar — *B.Sc. Thesis Supervisor*

Professor, Department of EEE, Islamic University of Technology (IUT)

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